



Society of Petroleum Engineers

SPE-229891-MS

Impact of Low Concentration Ferric Chloride on Physical and Electrical Parameters in Low Permeable Limestone Formation

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229891-MS](https://doi.org/10.2118/229891-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Variation in rock type or any physical alterations in the rock has great impact on the Archie's parameter, unlike the brine conductivity which has no influence on these parameters. When the Ferric chloride (FeCl_3) solution is injected into limestone formation, it reacts with the carbonates and alters the permeability. The current study aims to investigate the impact of low concentration FeCl_3 on physical and electrical properties by intentionally altering the rock's characteristics which in turns changes the resistivity. The study uses Indiana limestone core samples having length of 2 inches and diameter of 1.5 inches. The centrifugation method was used to achieve uniform FeCl_3 saturation levels inside the core samples while investigating the variation in their physical and electrical properties. The resistivity of core samples were determined at different FeCl_3 saturations to examine its effect on the electrical properties. Medical CT was conducted on the core sample, both before and after treatment in order to understand the structure variation during the saturation and De-saturation. Furthermore, NMR analysis was performed for comprehensive insight into the pore system and fluid distribution inside the core sample. Additionally, geomechanical analysis showed a moderate decrease in Young's modulus and a slight increase in Poisson's ratio, consistent with mild matrix weakening.

The initial porosity and permeability of samples were measured to be 17.7% and 16 mD respectively. The results indicates that the Archie equation can be modified for fluid-rock interaction, providing a better understanding of the resistivity changes in the limestone formation. The findings also showed the stimulation of the limestone reservoir due to the acidizing effect of FeCl_3 . The FeCl_3 on interacting with water forms HCl which is responsible for dissolving the carbonates present in the limestone samples developing wormhole across the core sample, enabling the enhanced fluid flow through the rock. The stimulation of rock can be quantified by resistivity changes and enabling precise estimation of Archie parameters. This stimulation also causes a variation in the porosity and permeability of the core samples. The

NMR results suggests that the contribution of macro pores in total porosity has shifted towards the micro pores. This research highlights the impact of FeCl_3 interactions when applying Archie's Law in reservoir engineering. FeCl_3 in low concentration can be used as acidizing agent in limestone formation and enhancing its permeability for improved hydrocarbon recovery. The experimental results provide a pathway for refining Archie parameters in limestone formations undergoing stimulation and the individual pore contribution to the porosity shifts from a large pore system to small pore system.

Introduction

In reservoir characterization it is important to understand the relationship that is present between fluid behavior and the rock properties in order to correctly interpret the formation evaluation data. Archie's Law (Archie 1942) is the most basic empirical model of the correlation between the porosity of a rock and its electrical resistivity and is used to obtain estimates of water saturation so that reserves of hydrocarbons can be calculated [1]. In clean sandstones, Archie gives a good correlation through the cementation exponent "m" and through the formation factor, however it tends to cause large deviations in case of carbonates due to its vuggy porosity and heterogenous pore networks [2]. The significance of comprehending these constraints is that any chemical change of the pore structure such as acid stimulation will have additional impacts on the Archie's parameters and formation factor of carbonates rocks.

A traditional method of improving the carbonate reservoir permeability is through hydraulic fracturing where large amount of fracturing fluid is pumped at a pressure above fracturing pressure [3,4]. Another method includes the dissolution of a part of the rock using acid known to be called matrix acidizing. During the treatment, an acid is injected under nonfracturing conditions to dissolve minerals, expanding existing pore channels, or opening new ones (so-called wormholes) that can bypass near-wellbore damage [5]. Since the 1890s, hydrochloric acid (HCl) has remained the most common acid of choice in carbonate stimulation because it is very effective and it reacts with calcite and dolomite at a high rate [6]. However, it has some drawbacks including corrositivity and insufficient penetration [7]. HCl reacts violently with limestone, dissolving calcium carbonate to yield calcium chloride, carbon dioxide and water; in some circumstances this reaction can produce extremely conductive wormhole pathways that open up a vastly increased flow potential in otherwise low-permeability areas. In the matrix acidizing, single and dominant wormholes production is favored. In this line of research, prior experimental work indicated that there is an optimal rate of acid injection at which the greatest wormholing efficiency is generated [8–10]. In addition, high concentrations of HCl are very corrosive and require a lot of corrosion inhibitors and specialized materials, and may lead to high cost drilling, acid volume need, and uneven stimulation in mixed rock. These limitations encourage the invention of retarded acid systems and other methods of acids to obtain a better depth and evenness of stimulation.

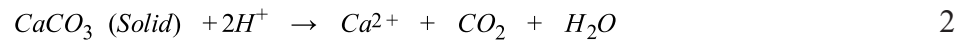
Numerous other acidizing solutions are investigated to address the drawbacks of HCl, including organic acids, emulsified acids and chelating agents [11–13]. The commonality of these engineered systems is that they are intended to keep the acid reactivity under control, slower, in order to expand the pore channels in a more controlled manner and to spill acid into areas that are less permeable. At this, ferric chloride (FeCl_3) turns out a rather exotic yet interesting option as a controlled acidizing agent. When Ferric chloride is dissolved in water, the chloride salt is itself not hydrolyzed, rather it acts as a source of HCl which is generated in situ, giving a strongly acidic solution (pH ~2) through hydrolysis to form hydrochloric acid and ferric hydroxide [14]. It follows that an FeCl_3 solution has the potential to become a self-acidizing fluid and the HCl produced will gradually dissolve carbonate minerals during its invasion of the formation. More importantly, the by-products of reaction consist of solid iron (III) hydroxides that precipitate immediately the local acid is exhausted (when pH reaches a value greater than ~2 to 3) [15]. This dissolution vs precipitation process in ferric-chloride-acidized carbonate cores was demonstrated by Rege and Fogler (1989) where the injected FeCl_3 solution did indeed form worm hole like channels, but it showed periodic drops in

permeability as $\text{Fe}(\text{OH})_3$ was formed that would clog up and block the flow, re-routing the pattern of acid flow [15]. This self-diverting behavior gives the impression that under appropriate control conditions FeCl_3 would potentially induce a more homogeneous expansion of pore space in low-permeability carbonates.

- Hydrolysis of FeCl_3 in water



- Acid Dissolution of Calcite



Although ferric chloride, or FeCl_3 , was correctly applied in the given study as a stimulation agent, it could also enter subsurface formations during regular oilfield activity. FeCl_3 in drilling has been used to inhibit shale and as flocculating agent in water-based mud systems to prevent clay swelling and separation of solids [16]. During workover and completion operations, Fe^{3+} may be deposited in brines as a corrosion product when steel tubing or casing is contacted with oxygenated fluids (in particular, acidic or high-salinity conditions). The presence of Fe^{3+} arising corrosion of steel tubulars in oxygenated brines in completion and workover fluids may create the possibility of eliciting ferric chloride in effluents. When acidizing, the corrosion of iron scales or metal surfaces by HCl may cause oxidation of Fe^{2+} to Fe^{3+} , which forms FeCl_3 in situ; when the pH rises ferric ions will have a tendency to hydrolyze and precipitate, clogging pore throats or forming sludge when added to crude oil surfaces, impairing formation permeability (well-documented experimentally with FeCl_3 systems) [15].

It is important to identify such sources of FeCl_3 that are generated operationally, since small concentrations can have dramatic impacts on pore structure, resistivity and formation evaluation responses. Therefore, the study tries to replicate such conditions experimentally to evaluate the petrophysical and electrical influence of FeCl_3 on carbonate rocks. The objective of the current research is the assessment of the effects of a low concentration solution of FeCl_3 on the electrical and physical characteristics of low permeability limestone, where Indiana Limestone is the tested sample material. The knowledge gained in this work will provide insights into the viability of the FeCl_3 controlled wormholing agent, as well as, how such a chemical transformation can be considered in the context of formation evaluation. Finally, this piece of work gives a source of upgrading stimulation techniques in low yield carbonate formations and updating petrophysical deductions (e.g. calculation of saturation) in acidizing treated reservoirs.

Methodology

This experiment was conducted in a well-designed experimental procedure to assess the petrophysical response of core samples under different conditions. The exercise comprised dry and saturated resistivity, acoustic measurements, then drainage and imbibition to mimic reservoir conditions with the aid of a centrifuge. The stages were set to record important parameters, e.g., properties depending on saturation. The series offers an understanding of fluid displacement and rock-fluid interactions as shown in Figure 1.

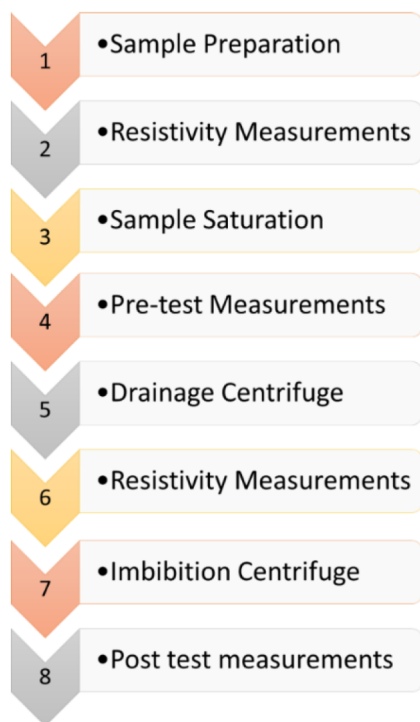


Figure 1—Workflow for the experimental procedures

Sample and Fluid Preparation

Indiana limestone core plugs (outcrop carbonate) of 2 inches length and 1.5 inches diameter were used for all experiments, having porosity 17.7 % and permeability 16.6 mD. All the core plugs were subjected to a uniform cleaning and drying process in order to have a clean and reproducible starting-condition. The saturating fluid was a ferric chloride (FeCl_3) brine doped into a potassium chloride (KCl) base brine. Particularly, a 3% KCl solution was prepared using deionized water and then 10% by weight of anhydrous FeCl_3 was dissolved in the solution. The KCl brine was used as a stabilizing solution and was chosen because of its widespread application in oil field core analyses as well its moderate conductivity which allows it to contrast with more reactive fluids. The ferric chloride was slowly added with continuous stirring so that local supersaturation and precipitation did not occur due to mixing. The resultant solution was then poured through a membrane to sieve off any suspended solids or colloidal iron hydroxides. The resistivity of the fluid were determined as shown in [table 1](#).

Table 1—Resistivity of the fluids

Solution	Resistance (ohm)	Resistivity (ohm-m)
KCl	16.56	0.349
FeCl_3	4.46	0.094

Characterization of Samples

Characterization of the limestone core samples was done under various physical, electrical and mechanical domains to determine the changes caused by stimulating agent (FeCl_3). All these characterization methods can be summed up as given below:

- **Petrophysical Properties:** The effective porosity was identified through helium porosimetry, which laid down a baseline on fluid transport capacity. High-resolution medical-grade Medical CT images in both 2D and 3D formats were collected to be able to visualize the internal pore structure and mineral distribution. Further, Nuclear Magnetic Resonance (NMR) spectroscopy was used to evaluate the fluid distribution in the pore network and the relaxation time of T_2 was used to determine the macro- vs. micro-pore distribution and pore structure development following treatment.
- **Electrical Parameters:** Measurements of resistivity were done to establish formation resistivity at fully saturated conditions by Vinci EPS-A system. The repeated measurements were conducted after the treatment in order to observe the effect of FeCl_3 on electrical conductivity and the evaluation of difference in Archie parameters such as formation factor (a), cementation exponent (m) and saturation exponent (n).
- **Acoustic Measurements:** Acoustic velocity tests were done under confining pressure (1500 psi) to give the emulation of subsurface stress. Compressional wave velocity V_p was measured with a pulse-transmission ultrasonic testing. These values were adopted to obtain the Young modulus and the Poisson ratio, which provided an idea about the frame stiffness of the rock and the effect of chemical stimulation on its structure.

Results and Discussion

This section provides a comparative report of the physical, electrical and geomechanical tests performed on the core samples of Indiana limestone before and after treatment. The results demonstrate the changes of the stimulation of the pore structure by FeCl_3 , effects on fluid flow properties and resistivity behaviors that have been supplemented by porosity, resistivity, NMR, CT, and acoustic tests.

Porosity and Pore Structure Alteration

The porosity of the limestone core was 17.7 % which increased to 19.2 % due to treatment with FeCl_3 solution. This ~1.5-unit increase in porosity is evidence that the FeCl_3 stimulation procedure produced more pore volume, probably by dissolution of carbonate matrix. The well-known concept of acidizing carbonate formations is increasing porosity by dissolving of minerals and by creating wormholes [17]. In our case, even a low concentration of FeCl_3 produced measurable matrix removal. It is worth noting that the preparation of FeCl_3 may have precipitation influence, since ferric iron is hydrolysed, some pore throats may be blocked by ferric hydroxide precipitation [15]. This type of textural changes is also in line with literature that pore architecture complexity can be changed with acid stimulations. The water saturation levels of the rock samples were calculated during the centrifuge (table 2), followed by viscosity determination of KCl and FeCl_3 solutions to be 1.04 and 1.29 cP, respectively.

Table 2—Saturations of rock samples during centrifuge

Sample	Saturation of rock sample	
	After Drainage	After Imbibition
1	22.4 %	73 %
2	39.4 %	80 %

Electrical Resistivity and Archie's Parameters

The resistivity of the core presented a severe reduction as a result of saturation with the solution of FeCl_3 . The resistivity of the rock sample saturated with 3 % KCl brine (sample 1) was 8.593 ohm-m, and FeCl_3 saturated (sample 2) lowered the resistivity to ~3.215 ohm-m (100 percent of brine saturation). This substantial

reduction is anticipated, as the ionic strength of the FeCl_3 solution is larger than that of the KCl brine. All the resistivity values has been calculated in [table 3](#).

Table 3—Resistivity of the rock samples at different saturations

Sample		Resistance (ohm)	Resistivity (ohm.m)
No.	Type		
1	Dry	4.63×10^7	10.3×10^8
1	KCl saturated	385	8.593
1	After Drainage	2.69×10^4	602.64
1	After Imbibition	561.36	12.55
2	Dry	6.91×10^7	15.4×10^8
2	KCl saturated	144	3.215
2	After Drainage	2670	59.68
2	After Imbibition	275.06	6.14

In addition to fluid conductivity effects, electrical behavior was also attributed to changes in pore-structure. We recomputed the parameters of Archie, before and after treatment as in [table 4](#). The overall implication of these changes in the Archie parameters is that the pore network was made less electrically efficient - i.e. more tortuous or poorly connected due to FeCl_3 treatment.

Table 4—Archie's parameters of the rock samples

Sample	a (tortuosity factor)	m (cementation exponent)	n (saturation exponent)
1 (KCl saturated)	1.036	1.49	2
2 (FeCl_3 saturated)	1.2	1.677	2

Nuclear Magnetic Resonance and Medical Computed Tomography Analysis

Nuclear Magnetic Resonance (NMR) T_2 relaxation measurements carried out before and after treatment demonstrated a prominent change in distribution of pore sizes. Initially, the limestone cores were dominated by macro-porous nature with increased relaxation time, and after treatment the T_2 distribution moved to shorter times, hence, the onset of micro to mesoporosity. Such transition implies that the acidizing process created smaller pores or etched existing pores surface, thereby creating more internal surface area and complexity. The results can be visualized from [figure 2](#). The related changes of NMR T_2 profiles have been described quite well in acidized carbonate reservoirs. As an example, Jora et al. (2021) showed that pore size redistribution towards smaller features can also be observed in the case of static acid dissolution, which is possible to recognize by NMR and validated with the extra power of CT imaging [18]. Similarly, Wang et al. (2020) found that acidification of the mixed siliciclastic-carbonate cores resulted in a decreased T_2 relaxation time, which is commonly shown due to narrowing of the pore throats and altering the structures [19].

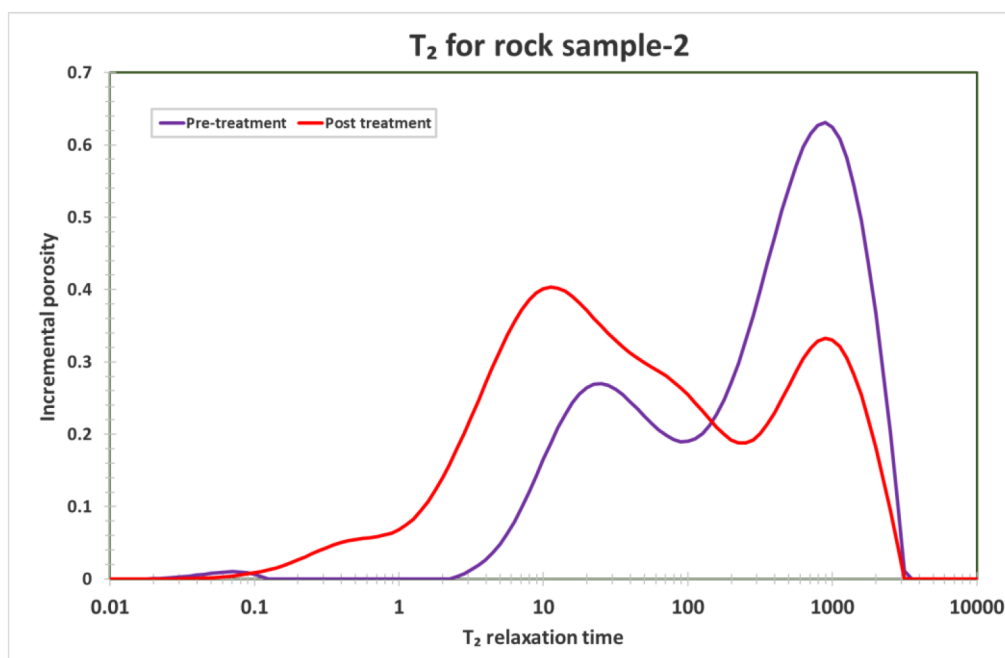


Figure 2—NMR analysis for rock sample (FeCl_3 saturated)

Internal rock structure before and after treatment was also visualized by use of Medical computed tomography (CT). 2D and 3D medical CT images at a high resolution showed some textural variations after acidization, and the concept of wormhole-like pathways and uneven density areas was observed (figure 3). This is visual confirmation of the trends in porosity and resistivity and proves that FeCl_3 produced channels of dissolution and modifications to the fabric of the whole. Micro-CT was employed by Yoo et al. (2021) who also described wormhole propagation in carbonates and found elongated dissolution features that were compatible with acid-rock reaction and the ability of medical CT to map the acid-induced microstructural changes in carbonate rocks made by CT was highlighted in the research by Icuero Urgel-Pinto and Alcazar-Vara (2024) [20,21].

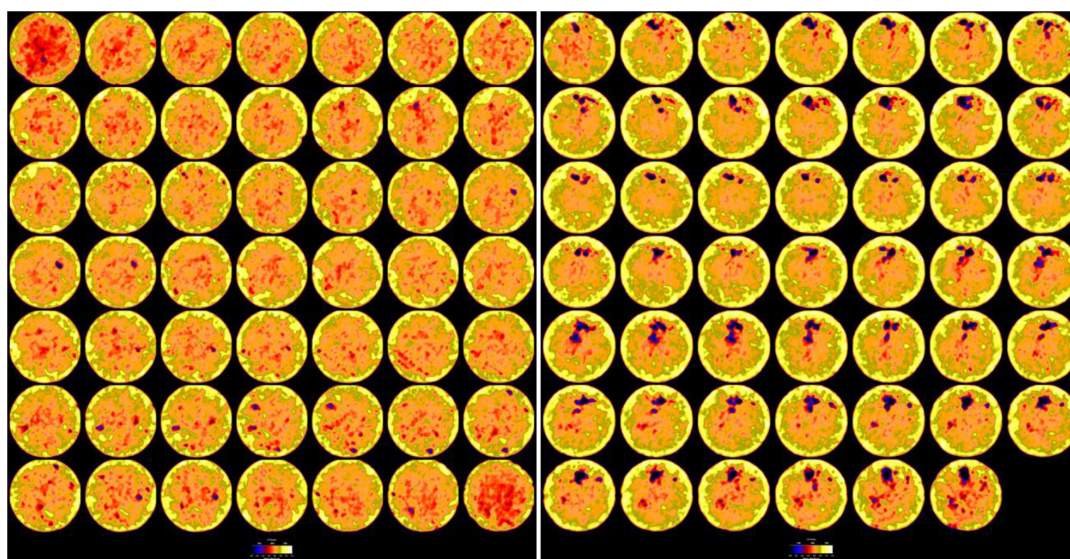


Figure 3—2D imaging using medical CT for rock sample (FeCl_3 saturated)

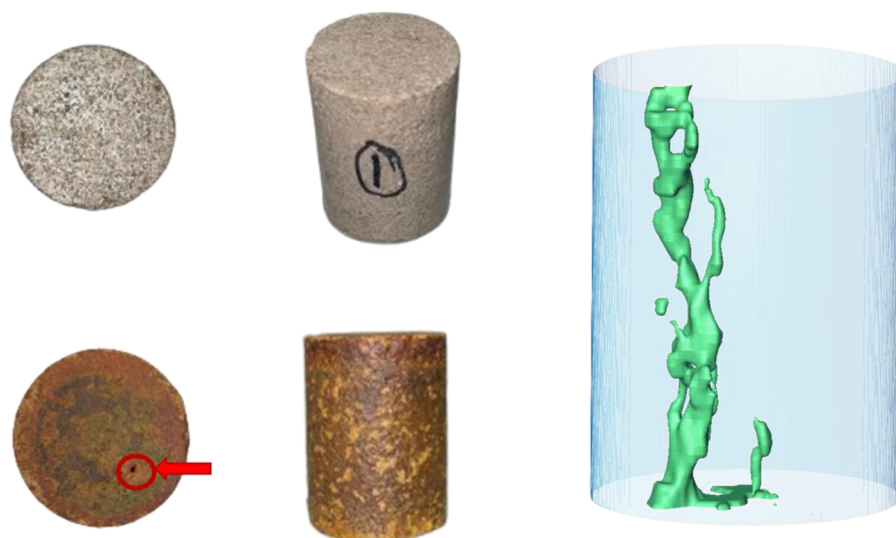


Figure 4—3D imaging using medical CT for rock sample (FeCl_3 saturated)

Geomechanical Analysis

In order to assess the ultrasonic acoustic testing of FeCl_3 impact to the mechanical integrity of limestone, a pulse transmission method testing was done under a confining pressure of 1500 psi. This compressional wave velocity (V_p) and shear wave (V_s) was measured, followed by calculating the elastic moduli (Young modulus and Poisson ratio). A decrease in Young's modulus and a concomitant increase in Poisson ratio of a small amount after chemical stim could be said to mean that there is a slight decrease in stiffness and ductility. These mechanical reactions are in line with the published acid-rock interactions in the literature. Alameedy et al. (2022) confirmed similar changes in acid-treated limestone cores and showed 26-37% reduction in Young modulus and increased Poisson's ratio due to weakening of the grain boundaries and microcracks caused by the acid [17]. Similarly, Parandeh et al. (2023) established that HCl-based remedies greatly decrease the mechanical strength of carbonated rock as a result of the enlargement of pores and corrosion of the matrix [22].

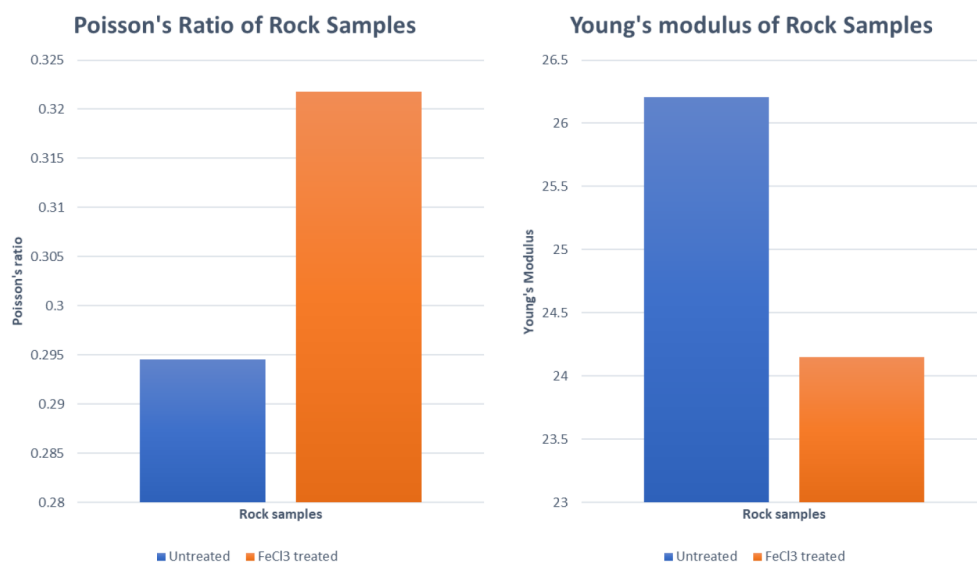


Figure 5—Geomechanical properties for rock sample (FeCl_3 saturated)

Conclusion

The research establishes the fact that with the combination of the petrophysical, electrical, and geomechanical properties, it can be shown that low-concentration FeCl_3 successfully modifies the internal formation and conductivity of tight limestone rock. The main observations are the following:

- Porosity increased from 17.7 % to 19.2 %
- The fluid conductivity and pore network evolution contributed to a decrease in electrical resistivity.
- Archie parameters (a , m) increased, indicating more tortuous microstructure.
- NMR and CT verified that there was a transition towards finer pores and the development of etched pathways.
- Mechanical stiffness decreased modestly, consistent with controlled matrix softening.

These results validate more wide ranging conclusions obtained in acidizing studies. A recent review by Correia da Silva et al. (2024) pointed out that carbonate acidizing does not increase the porosity and permeability alone but also produces observable changes in the mechanical and electrical properties - where all of them have to be addressed in reservoir modeling [23]. Alhamad et al. (2020) also applauded the potential of alternative acid systems such as the use of organic acids or iron-based acids in view of controlled, deeper stimulation in the context of reduced near-wellbore damage [24]. Taken together, our findings indicate that FeCl_3 can be used as a stimulation fluid in low-permeability carbonates. It increases the flow capacity and alters petrophysical responses without harshly compromising structural stability, which is essential when it comes to the field-level application.

Acknowledgement

The authors would like to acknowledge the College of Petroleum and Geosciences, King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia, for providing the facilities and funding support.

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